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Department of Computer Engineering and Applications
Institute of Engineering & Technology

# BCSE-1501
# Network Security

Unit-3

Prepared By: MS. Priyanka Sharma
Email : priyankasharma.1@gla.ac.in

# Internet Protocol Security

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IPsec is a security method used to protect data sent over the Internet.
It makes sure that the information is safe, private, and not changed while traveling from one device to another.

**Purpose**

- IPsec encrypts data so outsiders cannot read it.
- It authenticates devices, ensuring data comes from a trusted source.

**How IPsec Works**

- It works at the Network Layer, meaning it protects data before it goes onto the Internet.
- It is commonly used to create Virtual Private Networks (VPNs).

**Modes of IPsec**

- Transport Mode
  - Encrypts only the data part of the packet.
  - Used mainly for device-to-device communication.
- Tunnel Mode: Encryptes the entire IP Packet.
- Used for VPN Connections between networks and users.

# Transport Mode

- Transport Mode protects only the actual data being sent, not the address information.
- Only the data (payload) is encrypted
  → The message is hidden, but the address stays visible.
- IP header remains visible
  → Routers and network devices can still see where to send the packet.
- Used for device-to-device communication
  → Common when two computers or servers communicate securely with each other.

# Tunnel Mode

**Tunnel Mode** protects **everything** in the data packet.

- **Entire IP packet is encrypted**
  → Both the **data and the original address information** are hidden.
- **New IP packet is created**
  → The original packet is placed inside a new one with a new header.

**Used for network-to-network security**
→ Common in **VPNs**, such as connecting two office networks or a user to a company network.



# Key Components of IPsec

**Security Association (SA)**

- An **SA is like a security agreement** between two devices.
- It decides **how data will be protected**, such as:
  - Which encryption method to use
  - Which keys are used
- Each secure connection has its **own SA**.

**Internet Key Exchange (IKE)**

- **IKE sets up the secure connection automatically**.
- It:
  - Creates and manages Security Associations
  - Safely exchanges encryption keys
- No manual key sharing is needed.

**Cryptographic Algorithms**

- Encryption: AES, 3DES
- Integrity: HMAC-SHA1, HMAC-SHA256
- Authentication: RSA, ECDSA, Pre-Shared Key (PSK)

# Tools for Managing HTTPS



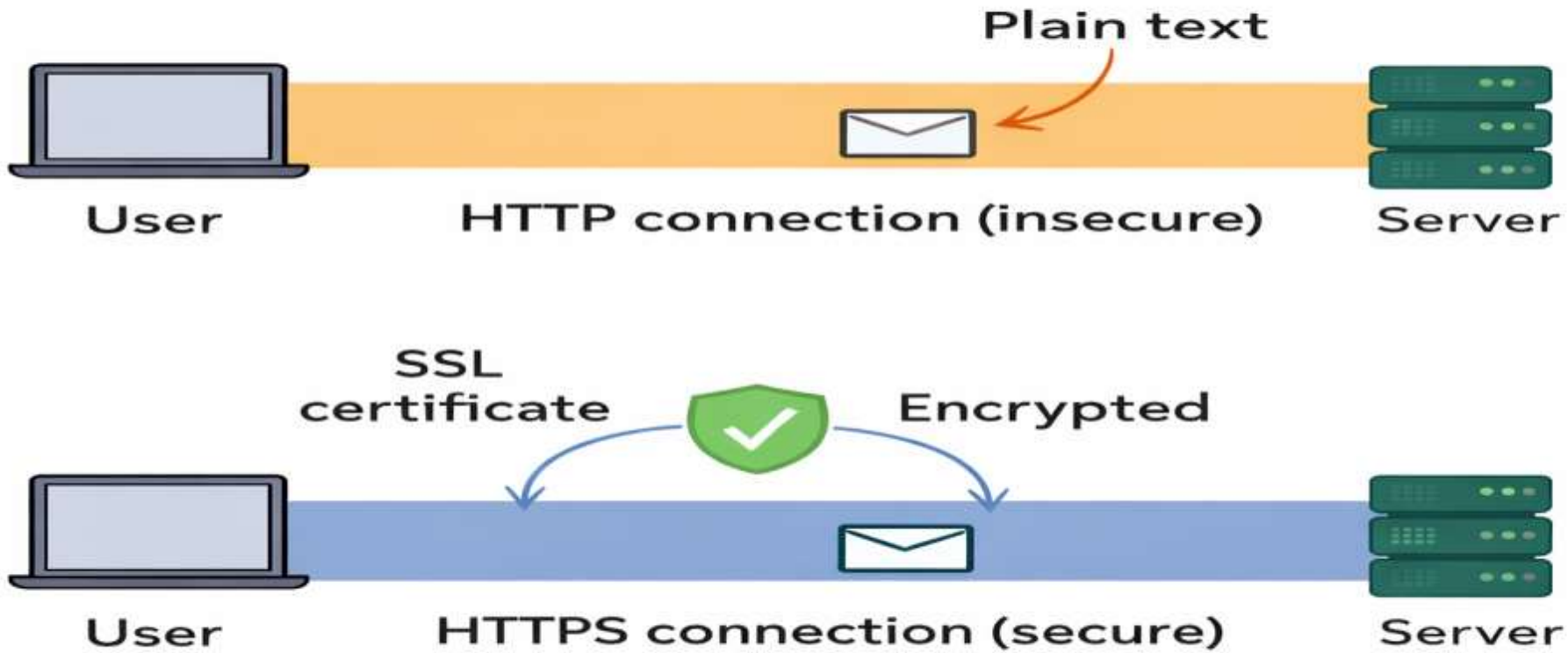
**1. Certificate Authorities (CAs)**

- Certificate Authorities are trusted organizations.
- They **issue SSL/TLS certificates** to websites.
- These certificates prove that a website is **real and secure**.
- Examples:
  - **Let's Encrypt** – Free certificates
  - **DigiCert** – Paid, widely used by companies
  - **GlobalSign** – Trusted for business websites

**2. SSL/TLS Testing Tools**

- These tools **check whether HTTPS is configured correctly** on a website.
- They find **security problems** like weak encryption or old TLS versions.
- They also give **suggestions to improve security**.
- Example:
  - **SSL Labs** – Tests SSL/TLS strength and gives a security grade

# HTTP vs HTTPS



# HTTP + SSL = HTTPS

**(Hypertext Transfer Protocol)**

*Defines how messages are transmitted between visitor's browser and website's server.*

**(Secure Socket Layer)**

*Protects and encrypts information sent across the Internet.*

**(Hypertext Transfer Protocol Secure)**

*Encrypts information sent between browser and server.*

HTTPS makes it harder for hackers to break the connection and steal personal information such as credit card numbers, addresses, passwords, etc.

**HTTPS helps further protect the privacy of your visitors!**

# Secure Shell (SSH)

**Secure Shell (SSH)** is a network protocol used to **connect to another computer securely**.

- It works over **untrusted networks** like the internet.
- SSH is mainly used for:
  - **Remote login** (accessing servers from another system)
  - **Secure file transfer**
  - **Running commands on servers**

**Why SSH is Secure?**

- **Encryption** – Data is hidden, so hackers cannot read it.
- **Authentication** – Confirms the user and server are genuine.
- **Integrity** – Ensures data is not changed during transfer.

# Working of SSH



**Connection Establishment**

- The **SSH client** (user's computer) connects to the **SSH server**.
- By default, SSH uses **Port 22**.

**Server Authentication**

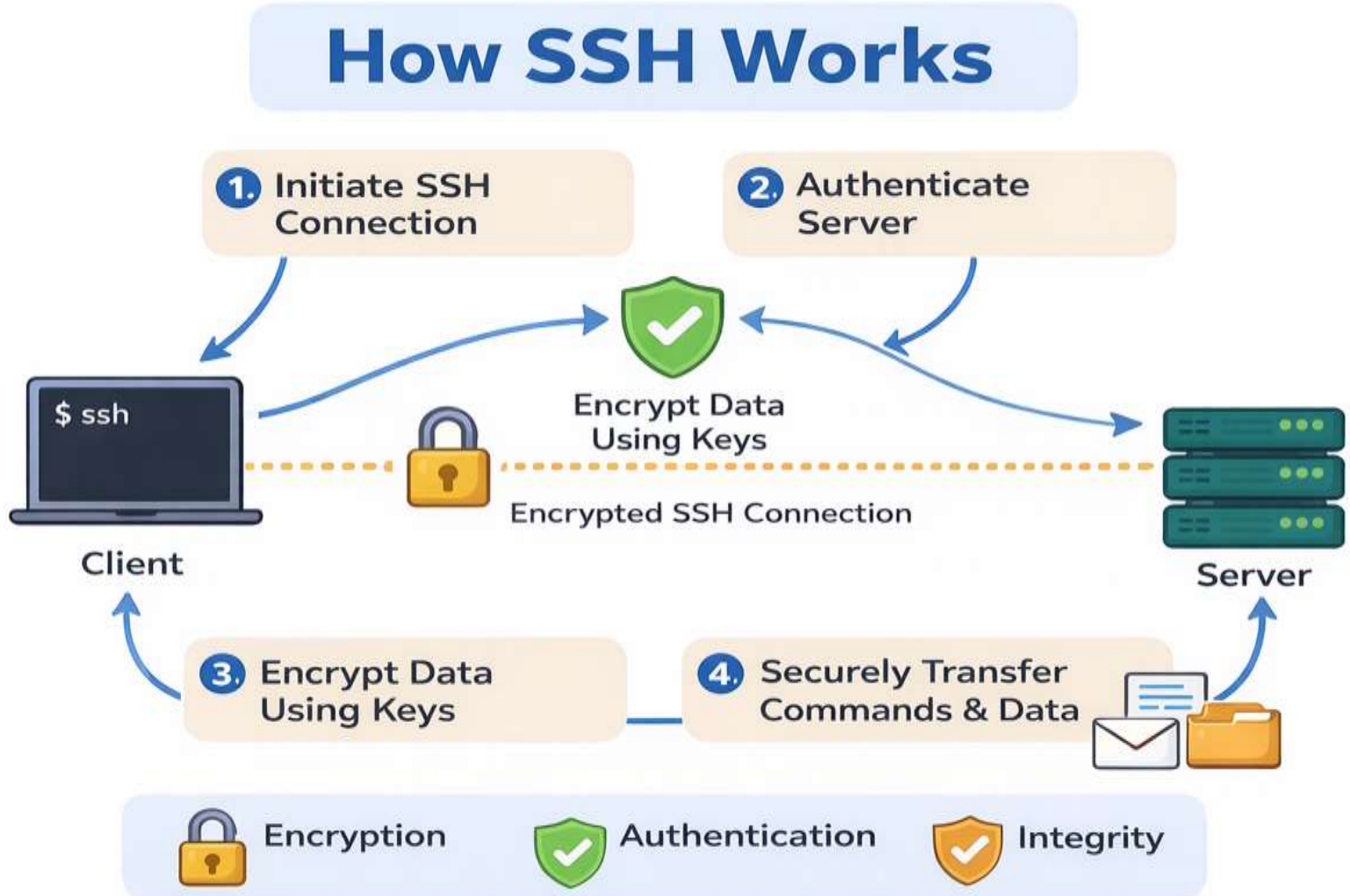
- The server sends its **public key** to the client.
- The client checks this key to make sure the server is **genuine**.
- This prevents fake or malicious servers. **Key Exchange**
- Both client and server use a **key exchange algorithm** (like Diffie-Hellman).
- A **shared secret session key** is created.
- This key is used to **encrypt all communication**.

**User Authentication**

- The user proves their identity using:
- Password authentication
  **OR**
- Public & private key authentication (more secure)

**Secure Communication**

- After authentication, a **secure encrypted tunnel** is created.
- All commands, files, and data are **encrypted**.
- Hackers cannot read or modify the data.


How SSH Works
1. Initiate SSH Connection
2. Authenticate Server
$ ssh
Encrypt Data Using Keys
Encrypted SSH Connection
Client
Server
3. Encrypt Data Using Keys
4. Securely Transfer Commands & Data
Encryption
Authentication
Integrity

# Components of SSH (Secure Shell)

**1.SSH Client**

- The **SSH Client** is the user's computer or software.
- It is used to **start the SSH connection**.
- Examples: **OpenSSH, SecureCRT**

**2.SSH Server**

- The **SSH Server** runs on the remote machine.
- It **accepts SSH connections** from clients.
- Listens on **Port 22** by default.
- Example: **OpenSSH Server**

**3.SSH Daemon (sshd)**

- A background program running on the server.
- Manages:
  - User authentication
  - Secure session setup
  - Encrypted communication

# Components of SSH (Secure Shell)



**Authentication Mechanism**

- Used to verify the user's identity:
- Password Authentication
- Public Key Authentication
- Certificate-based Authentication
- Two-Factor Authentication (2FA)

**Key Exchange Mechanism**

- Used to create a secure session key.
- Ensures both sides agree on encryption keys.
- Example algorithms:
  - Diffie-Hellman
  - Elliptic Curve Diffie-Hellman (ECDH**)**

**Encryption Algorithms**

- Protect data from being read by attackers.
- Common algorithms:
  - **AES**
  - **ChaCha20**

# Components of SSH (Secure Shell)

**Message Authentication Code (MAC)**

- Ensures **data integrity**.
- Detects any changes during transmission.
- Examples:
  - **HMAC-SHA2**
  - **Poly1305**

**Secure Channel**

- An **encrypted tunnel** created after authentication.
- Used to transfer:
  - Commands
  - Files (SCP / SFTP)
  - Terminal data